Digital CAD Training and Services

CADPOINT
AUTHORIZED TRAINING CENTRE
ISO 9001: 2008 CERTIFIED COMPANY

CATIA V5 45 Days Syllabus

Day-wise learning plan with practical tasks, project work and final assessment

Duration	Software	Training Focus
45 Days	**CATIA V5**	Sketcher, Part Design, Wireframe, Surface, Assembly, Drafting, Project

This document is prepared for students to clearly understand the complete course journey, daily topics, practice expectations, project work and final outcome before joining the training.

Course Objective

This syllabus is designed to build a strong practical foundation in CATIA V5 through a clear 45-day progression. Students begin with Sketcher fundamentals, move into Part Design, continue with Wireframe and Surface tools, learn Assembly and Drafting workflows, and complete the course with project work and final drawing assessment.

What Students Will Learn

- Create fully constrained sketches using CATIA Sketcher tools and constraints.
- Build 3D parts using Pad, Pocket, Hole, Fillet, Chamfer, Draft, Shell, Rib, Slot, Pattern and Multi Section Solid.
- Create reference geometry using points, planes, lines and axes.
- Create and edit surface features such as Extrude, Revolve, Sweep, Multi Section Surface, Boundary, Join, Trim, Split, Fillet and Offset.
- Prepare assemblies using CATIA constraints and exploded views.
- Create production drawings with views, sections, dimensions, tolerances, GD&T, BOM, balloons and title block setup.

Module Breakdown

Module	Days	Focus Area
Sketcher	**5**	2D profile creation, constraints, dimensions and sketch analysis
Part Design	**10**	Core 3D feature creation, editing and feature management
Wireframe	**2**	Reference geometry creation using points, planes, lines and axes
Surface	**11**	Surface creation, trimming, joining, splitting, offset and analysis
Assembly	**5**	Component insertion, constraints and exploded view creation
Drafting	**9**	Drawing setup, views, dimensions, tolerances, GD&T, BOM and templates
Project	**3**	Mini project execution, final drawing preparation and assessment

Complete Day-wise Syllabus

Each day includes one focused topic and one daily practice task. Students should complete the practice task on the same day to maintain learning continuity and improve software command.

Day	Module	Topic	Daily Practice Task
1	Sketcher	CATIA Interface + Workbench Overview	Create fully constrained 2D plate profile
2	Sketcher	Sketch Tools - Line, Circle, Arc	Create bracket profile
3	Sketcher	Geometrical Constraints	Apply constraints to support profile
4	Sketcher	Dimensional Constraints	Modify sketch using trim/extend
5	Sketcher	Sketch Analysis	Create complex constrained sketch
6	Part Design	Pad Command	Create base solid block
7	Part Design	Pocket Command	Create internal cavity
8	Part Design	Hole Command	Create different hole types
9	Part Design	Fillet & Chamfer	Apply fillet/chamfer
10	Part Design	Draft Command	Apply draft to faces
11	Part Design	Shell Command	Hollow solid block
12	Part Design	Rib Command	Create reinforcement rib
13	Part Design	Slot Command	Create guide-based slot
14	Part Design	Multi Section Solid	Create lofted solid
15	Part Design	Pattern	Create repeated features
16	Wireframe	Points & Planes	Create reference geometry
17	Wireframe	Lines & Axis	Create wireframe geometry
18	Surface	Extrude Surface	Create extruded surface
19	Surface	Revolve Surface	Create cylindrical surface
20	Surface	Sweep	Create sweep surface
21	Surface	Multi Section Surface	Create MSS
22	Surface	Boundary	Create boundary surface
23	Surface	Join	Join surfaces
24	Surface	Trim	Trim surfaces
25	Surface	Split	Split solid
26	Surface	Fillet Surface	Apply surface fillet
27	Surface	Offset Surface	Create offset
28	Surface	Surface Analysis	Check quality
29	Assembly	Assembly Overview	Assemble 2 parts
30	Assembly	Coincidence Constraint	Apply contact
31	Assembly	Offset Constraint	Apply offset
32	Assembly	Angle Constraint	Apply angle
33	Assembly	Exploded View	Create explode

Day	Module	Topic	Daily Practice Task
34	Drafting	Drafting Workbench	Generate views
35	Drafting	Section View	Create section
36	Drafting	Dimensioning	Add dimensions
37	Drafting	Tolerances	Apply tolerances
38	Drafting	GD&T	Apply GD&T
39	Drafting	BOM	Create BOM
40	Drafting	Sheet Setup	Prepare final sheet
41	Drafting	Title Block and Drawing Template	Create title block and save a reusable CATIA drawing template
42	Drafting	BOM and Balloons Review	Prepare BOM, add balloons and verify item numbering
43	Project	Mini Project - Part Design	Create one complete CATIA part using sketch and part design features
44	Project	Mini Project - Assembly	Assemble project parts, apply constraints and create exploded view
45	Project	Final Drawing and Assessment	Prepare final drawing sheets and complete final evaluation submission

Student Practice Guidelines

- Complete the daily practice task before moving to the next day.
- Keep every file properly named with day number, topic and your name.
- Maintain separate folders for Sketcher, Part Design, Wireframe, Surface, Assembly, Drafting and Project files.
- Revise previous commands weekly to improve speed, confidence and command retention.
- Submit doubts with screenshots, file name and the exact step where you are facing the issue.
- Complete the mini project and final assessment before claiming course completion.

Final Outcome After 45 Days

Area	Expected Outcome
Sketcher	Students can create and modify fully constrained 2D profiles.
Part Design	Students can build practical 3D parts with common production features.
Wireframe	Students can create reference geometry required for part and surface work.
Surface	Students can create, trim, join, split, offset and analyze CATIA surfaces.
Assembly	Students can insert components, apply constraints and create exploded views.
Drafting	Students can prepare drawings with views, dimensions, tolerances, GD&T, BOM and title blocks.

Area	Expected Outcome
Project	Students can complete a mini project and submit final drawing sheets for assessment.

For course details and batch updates, contact Digital CAD Training and Services.